

Power-Domain CSI Prediction for Power Allocation in 6G MIMO-OFDM Systems

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Abstract—Dynamic downlink power allocation in multiple-input multiple-output (MIMO-OFDM) systems is constrained by the need to reacquire channel state information (CSI) whenever resource-block (RB) transmit-power configurations change. Since CSI feedback depends on the transmit power used during acquisition, evaluating alternative power allocations requires repeated CSI measurements and feedback, introducing signaling overhead and latency. In this paper, we formulate power-domain CSI prediction as the task of inferring CSI feedback indicators at unobserved transmit-power levels from CSI acquired at a reference power. To address this problem, we propose a power-domain CSI Transformer (PDCT) that predicts rank indicator (RI), channel quality indicator (CQI), and precoding matrix indicator (PMI) feedback under varying RB transmit-power configurations. We further introduce a proximity cross-entropy (PCE) loss that incorporates ordinal class offsets into the training objective. Simulations using the Vienna 5G system-level simulator show that PDCT achieves low normalized mean absolute error (NMAE) across all CSI indicators, reduces transmission-rate loss relative to baseline methods, and performs inference in tens of microseconds per sequence of 100 CSI. These results demonstrate the feasibility of power-domain CSI prediction as an intermediate capability for future CSI-aware power-allocation frameworks.

Index Terms—CSI feedback indicators, Power allocation, Link adaptation, Transformer networks, 6G and beyond networks

I. INTRODUCTION

Downlink power allocation in MIMO-OFDM systems is fundamentally constrained by the need to reacquire channel state information (CSI) whenever resource block (RB) transmit-power configurations change. Power allocation methods typically assume that the required CSI is available at the point of allocation [1]–[3] but do not address the acquisition of CSI feedback under changing RB transmit-power configurations. Changing the transmit power of an RB alters the link conditions observed by the user equipment (UE) and causes previously reported CSI feedback to become outdated for link adaptation. Updating the CSI requires reacquisition at the new transmit power, which adds signaling overhead and latency.

CSI compression: To reduce the signaling overhead associated with repeated CSI acquisition, a line of research seeks to represent CSI more efficiently. Methods such as CsiNet [4], CSI-GPT [5], and subsequent Transformer-based CSI feedback frameworks [6]–[8] learn compact CSI representations that

can be accurately reconstructed at the BS. These methods do however still require each CSI to be measured and reported at the transmit-power levels of interest. As a result, practical systems such as 3GPP NR assume fixed RB transmit power during CSI acquisition [9], [10].

CSI prediction: An alternative line of research circumvents repeated CSI acquisition by predicting unknown CSI from related channel and propagation context observations. Existing works exploit correlations across the temporal, spatial, frequency, and power domains to improve channel estimation and CSI prediction performance. Transformer-based channel prediction [11] leverages self-attention to reformulate sequential CSI prediction over time as a parallel multi-token output task, reducing error accumulation. More recent work such as TPN [12] also leverages self-attention for both time- and frequency correlations, while LinFormer [13] replaces self-attention with a linear time-aware module to reduce inference complexity. Digital Twin Channel-Aided CSI Prediction [14] extrapolates CSI across space through a Transformer-CNN network and virtual propagation environment. CaFTRA [15] similarly learns space-frequency CSI correlations but to support fully feedback-free transmission. In particular, CaFTRA captures frequency correlations by self-attending to multiple CSI tokens sequenced along the RBs. Other works have investigated channel and channel-statistics estimation from power measurements [16], [17]. Across these works, learning-based models exploit structure and correlations in channel observations to improve prediction accuracy. Parallel prediction and linearization techniques are also used to improve latency, scalability, and inference efficiency. However, prediction of CSI feedback indicators across transmit-power configurations remains largely unexplored.

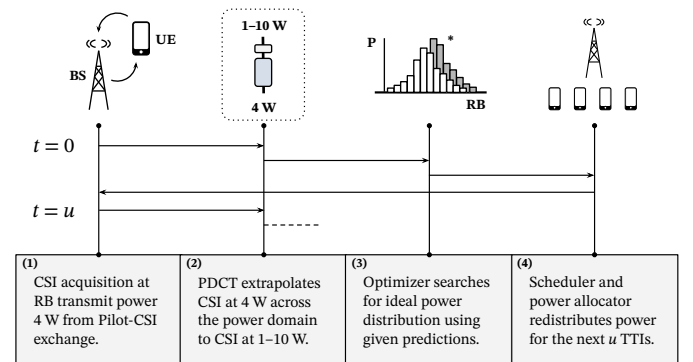


Fig. 1. Operational framework overview, with power-domain CSI Transformer (PDCT) at stage (2) shown as an intermediate power-allocation capability. Stages (1)–(4) are executed every u transmission time intervals (TTIs).

This paper: This work addresses power-domain CSI prediction as a missing capability between CSI acquisition and downlink power allocation. The objective is not to recover or extrapolate the physical channel, as considered in [11], [13], [14], [17], but to infer the CSI feedback indicators required for link adaptation under varying transmit-power configurations. We present a power-domain CSI Transformer (PDCT) that predicts CSI at a target transmit power from reference CSI and context. Within the operational framework shown in Fig. 1, the

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predicted CSI serves as an intermediate input to scheduling and power-allocation decisions, and remains available until the reference CSI is reacquired. This enables candidate RB transmit-power configurations to be evaluated without repeated CSI acquisition. In the simulations, CSI acquired at transmit power 4 W is used to predict CSI feedback at 1–10 W. The main contributions of this work are summarized as follows:

- 1) We formulate power-domain CSI prediction as a conditional learning problem to infer CSI feedback indicators at unobserved transmit-power levels and reduce the need for repeated acquisition under varying power.
- 2) We propose PDCT that exploits frequency-domain CSI structure and context for improved learning.
- 3) We extend cross-entropy (CE) loss to proximity cross-entropy (PCE) loss by penalizing ordinal class-offset errors and implementing wrap for periodic CSI fields.

II. SYSTEM MODEL

We consider downlink MIMO-OFDM transmission compliant with 3GPP TS 38.214 [10]. Following the 3GPP NR link-adaptation procedure, CSI is defined as the set of quantized channel-feedback indicators, consisting of one rank indicator (RI), two channel quality indicators (CQI), and one precoding matrix indicator (PMI) as,

$$\text{CSI} = \{\text{RI}, \text{CQI}_1, \text{CQI}_2, \text{PMI}\},$$

rather than a physical channel. RI indicates the recommended number of spatial transmission layers, CQI the supported modulation and coding scheme, and PMI the preferred precoding matrix from a predefined codebook. We index these indicators by $k \in \text{CSI}$, and denote the corresponding cardinalities as $C_k \in \{4, 16, 16, 72\}$. For modeling purposes, PMI is represented by its constituent codebook indices, beam azimuth i_{11} and elevation i_{13} . In the reported evaluation, i_{11} and i_{13} are combined to PMI. The UE derives the CSI feedback indicators from the signal-to-interference-plus-noise ratio (SINR), where RI and PMI are selected to maximize the post-equalization mutual information over the codebook, and CQI is set to the highest modulation-and-coding scheme that keeps the block error rate below 10%. Since SINR depends on RB transmit power, the reported indicators vary under different RB power configurations.

A. Problem Formulation

Let CSI_P denote the CSI feedback reported at transmit power P . Given CSI observed at a reference transmit power P_{ref} , the objective is to predict the CSI that would be reported at a target transmit power P_{tgt} under identical context $\mathcal{C}$. For notational convenience, we define

$$\mathbf{x} = \text{CSI}_{P_{\text{ref}}}, \quad \mathbf{x}' = \text{CSI}_{P_{\text{tgt}}}, \quad \tilde{\mathbf{x}} = T(\mathbf{x}, \mathbf{c}), \quad \mathbf{c} = (P_{\text{tgt}}, \mathcal{C}),$$

where $\mathbf{x}$ denotes reference CSI, $\mathbf{x}'$ target CSI, $\tilde{\mathbf{x}}$ predicted CSI, and $\mathbf{c}$ the conditioning variables. The objective is therefore to learn the mapping T , as

$$T: \mathbf{x} \times \mathbf{c} \mapsto \mathbf{x}',$$

for multiple possible P_{tgt} and fixed $\mathcal{C}$ in $\mathbf{c}$. This assumes access to $\mathcal{C}$ at inference in a low-mobility scenario.

III. THE PROPOSED SOLUTION: PDCT

This section proposes the power-domain CSI Transformer (PDCT), capable of learning precise mappings of multi-CSI inputs via self attention.

A. Model Architecture

The transformer encoder architecture, seen in Fig. 2, should be read bottom-up. Inputs $\mathbf{x}$ and conditions $\mathbf{c}$ are encoded, concatenated and then projected through an initial linear layer to a latent representation $\mathbf{z}$. Thereafter, $\mathbf{z}$ passes through the encoder body consisting of a sequence of N encoder blocks. N is a hyperparameter, where we default to $N = 1$. Each encoder block consists of one multi-head self-attention (MHSA) block and one feed forward network (FFN), both with skip connections. At the prediction head, the transformed latent $\tilde{\mathbf{z}}$ passes through an FFN without a skip connection (non-assisted mapping) and produces 5 logit outputs from the heads h_1 – h_5 , corresponding to RI, CQI_1 , CQI_2 , i_{11} , and i_{13} . Lastly, softmax is applied to the logit outputs, which are concatenated to form the predicted CSI $\tilde{\mathbf{x}}$. Rotary positional embeddings (RoPE) [18] are symbolized with four dots enclosed in square brackets, intended to mimic the 2D rotation matrix. This formulation is inspired by the sinusoidal positional encoding introduced in [19].

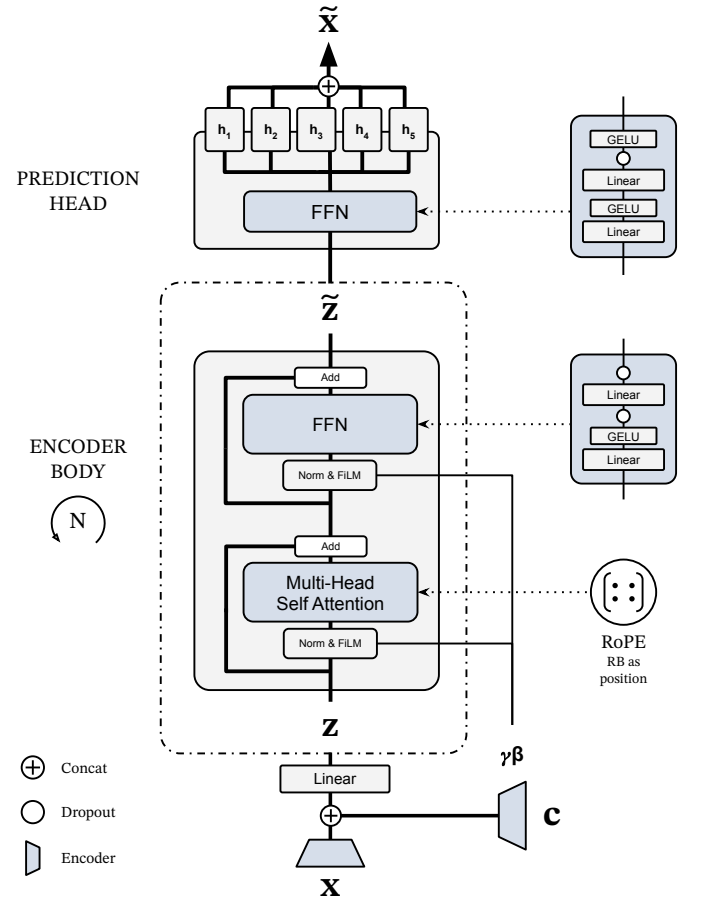


Fig. 2. Power-domain CSI Transformer (PDCT), predicting $\tilde{\mathbf{x}}$ ($\text{CSI}_{P_{\text{tgt}}}$) from $\mathbf{x}$ ($\text{CSI}_{P_{\text{ref}}}$) under conditions $\mathbf{c}$. Model instances are BS-native.

Encoders: The encoders learn latent representations for the inputs $\mathbf{x}$ and conditions $\mathbf{c}$. The CSI encoder maps $\mathbf{x}$ to learnable embeddings, concatenates them, and then projects the result through a linear layer. The condition encoder operates similarly but for $\mathbf{c}$. The linear projection layer then allows the encoder to learn in what way to mix the embeddings and best represent the encoded output.

Encoder Body: Feature-wise linear modulation (FiLM) [20] is used to scale and shift the latent as $\mathbf{z} \leftarrow \gamma \odot \mathbf{z} + \beta$, to make the conditioning more pronounced. Conditioning is crucial as the models need to learn multiple power mappings per reference power, only inferred by $\mathbf{c}$. The MHSA block then shares internal representations of CSI across tokens using scaled dot product attention (SDPA), similar to [11], [15] as,

$$\text{Attention}(Q_h, K_h, V_h) = \text{softmax}\left(\frac{Q_h^* K_h^{*\top}}{\sqrt{d_k}}\right) V_h.$$

CSI is correlated across the spatial, frequency, and power domains, allowing shared representations to promote structural understanding. SDPA is however order invariant¹ and does not encode distances between tokens in the aforementioned domains. With each input spanning all RBs of a BS, the RB index is used to encode positional awareness in the frequency domain via RoPE, where $*$ denotes rotated as,

$$\mathbf{x}_{\text{RB}}^* = \bigoplus_{i=1}^{d/2} R(\text{RB} \times \theta_i) \mathbf{x}_{\text{RB}}^{(i)}, \quad R_\theta = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}.$$

RoPE is applied to the queries Q and keys K (yielding Q^* and K^*), but not values V . The MHSA output forms a skip-connection with the latent before layer normalization and FiLM are applied. Skip connections enable the model to learn the residual mappings $h(x) = f(x) - x$, which facilitate the transformation of CSI into its target representation. The latent representation then passes through an analogous block in which MHSA is replaced by an FFN consisting of two linear layers interleaved with gaussian error linear unit (GELU) activations and dropout.

Prediction Head: After the encoder blocks, the transformed latent representation $\tilde{\mathbf{z}}$ is fed through an FFN, again consisting of two linear layers, GELU activations, and dropout. This FFN does not implement a skip connection, which forces the model to learn a full mapping $f(x)$ of the latent. Finally, the latent passes through 5 output heads h_1 – h_5 , each implemented as a linear layer outputting logits. Those logits are then decoded via argmax and lastly concatenated to form the transformed CSI $\tilde{\mathbf{x}}$.

B. Loss Function

The proposed method implements a custom loss, termed proximity cross entropy (PCE), which extends cross-entropy (CE) to account for ordinal prediction offsets. PCE uses distance-aware label smoothing, distributing probability mass around the target class, with mass decaying as the offset increases. This shifts the training objective from predicting the target class exactly to predicting as close as possible to

the target class, which benefits normalized mean absolute error (NMAE) scoring. PCE replaces the standard one-hot encoded targets with the rows of precomputed field-wise proximity matrices P_k , where

$$P_{k,ij} = \frac{\exp(-D_{k,ij}/\tau_k)}{\sum_{l=0}^{C_k-1} \exp(-D_{k,il}/\tau_k)}.$$

Here, τ_k is a temperature hyperparameter and $D_k \in \mathbb{R}^{C_k \times C_k}$ is a symmetric matrix encoding distances between the target class i and predicted class j . D_k encodes wrap for periodic fields, such as azimuth i_{11} , so that the offset is 1 between min and max ordinals as,

$$D_{k,ij} = \begin{cases} |i - j|, & \text{no wrap,} \\ \min(|i - j|, C_k - |i - j|), & \text{wrap.} \end{cases}$$

The proximity distribution, which replaces the one-hot target vector, is retrieved by indexing P_k with i . The temperature τ_k controls the spread of probability mass around the target class. A larger τ_k distributes more mass around the target class and is used for high-cardinality fields, such as CQI₁, CQI₂ and i_{11} , where prediction uncertainty is high. A smaller τ_k centralizes mass at the target class and is used for low-cardinality fields, such as RI and i_{13} , where accuracy should be emphasized.

C. Attention-Free Variant

Similar in motivation to LinFormer [13], which replaces self-attention with a linear module, we introduce PDCT-FFN, an attention-free variant of PDCT in which the MHSA block is replaced by a second FFN block in the encoder body. Simply removing the MHSA block would reduce the number of trainable parameters, making it difficult to distinguish between gains from SDPA and gains from increased model capacity. PDCT-FFN constitutes an alternative architecture within the proposed framework.

IV. EXPERIMENTAL SETUP

The communication environment was simulated using the Vienna 5G system-level simulator [21], employing a frequency-division duplex single-user MIMO configuration at 3.5 GHz with 100 MHz bandwidth divided into 500 RBs. Each BS was equipped with a single-polarized (6×1) antenna panel, while UEs employed four receive antennas and a 3GPP Type-I codebook. Channel propagation followed the geometry-based stochastic channel model specified in 3GPP TR 25.890, including large-scale path loss, shadowing, LoS/NLoS propagation states, and ITU Pedestrian-B small-scale fading. The simulation modeled an urban block of the Hong Kong Peng Cheng Laboratory, identical to [15]. The scenario was confined to a simulation space $\mathcal{S} = 400 \times 300 \times 75 \text{ m}^3$ with 5 BSs deployed on the rooftops of local buildings, shown in Fig. 3.

A. Simulation Data

For each BS, CSI and context data was sampled across 11,000 UEs, 100 RBs, and 10 RB power levels, forming 5 BS datasets of 11 million samples each. UEs were randomly positioned within $\mathcal{S}$ and power was quantized to discrete levels

¹SDPA cannot distinguish between permutations of the same set of tokens.

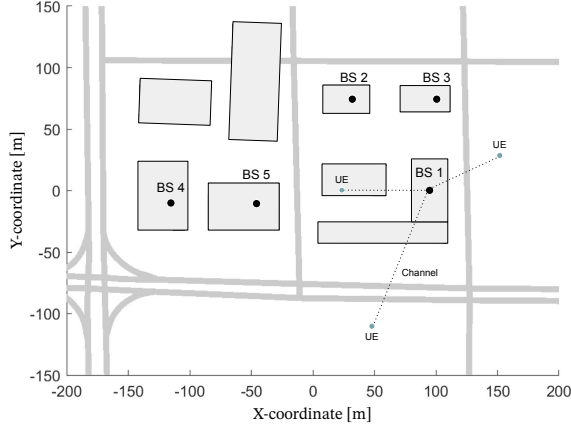


Fig. 3. 2D overview of the urban block (113.93371-113.93792, 22.57494-22.57709) from OpenStreetMap used in the Vienna 5G system-level simulations. The figure illustrates five BSs on the rooftops of nearby buildings and three example UEs.

1–10 W. Each sample consisted of 13 features: 5 CSI features and 8 context features. The CSI features were

$$\text{RI} \quad \text{CQI}_1 \quad \text{CQI}_2 \quad i_{11} \quad i_{13},$$

where azimuth, i_{11} , and elevation, i_{13} , were later used together with RI to form PMI when reporting the results in Section V. All 5 CSI features are ordinal and i_{11} is also periodic since it indexes the azimuthal beam direction. Therefore $i_{11} = 0$ and $i_{11} = 23$ are only $\frac{2\pi}{24}$ rad apart. The context features were

$$\mathbf{p}_{\text{BS}} \quad \mathbf{p}_{\text{UE}} \quad i_{\text{RB}} \quad P_{\text{RB}},$$

where $\mathbf{p} \in \mathcal{S}$ denotes the BS- and UE positions in 3D space. Since the simulation area was flat and we trained one model per BS, the UE and BS z-coordinates were constant. Each RB represented a contiguous group of OFDM subcarriers, indexed by i_{RB} , that defined its occupied frequency interval. P_{RB} was the quantized power allocated to the RB subcarriers, where higher power corresponded to higher achievable throughput.

B. Data Preparation

Three additional engineered features more explicitly modeling the downlink and beamforming geometry were added to each BS dataset. These were

$$\hat{\mathbf{u}}_\phi = (\cos \phi, \sin \phi) \quad \hat{\mathbf{u}}_{xyz} = \frac{\mathbf{d}_{xyz}}{r + \varepsilon} \quad r_{\log} = \log(1 + r)$$

where $\hat{\mathbf{u}}_\phi$ denotes the azimuthal direction, $\hat{\mathbf{u}}_{xyz}$ the LoS direction, and $r_{\log}$ the log BS–UE range. Here $\phi = \frac{2\pi}{24} i_{11}$ is the azimuthal angle, $\mathbf{d}_{xyz} = \mathbf{p}_{\text{BS}} - \mathbf{p}_{\text{UE}}$ the BS–UE Cartesian displacement vector, and $r = \|\mathbf{d}_{xyz}\|_2$ the BS–UE Euclidean distance. To avoid division by zero when normalizing $\mathbf{d}_{xyz}$, we used $\varepsilon = 10^{-8} \ll r$, not affecting the resulting geometry.

Note that $\hat{\mathbf{u}}_\phi$ and $\hat{\mathbf{u}}_{xyz}$ are distinct, despite encoding similar information. $\hat{\mathbf{u}}_\phi$ is the quantized direction that i_{11} encodes in the xy-plane, while $\hat{\mathbf{u}}_{xyz}$ is the exact LoS in 3D space. $r_{\log}$ encodes how free-space pathloss grows proportional to the log range. Importantly, there is not always a clear BS–UE LoS, meaning these features are not fully representative of the actual

transmission path. $\hat{\mathbf{u}}_\phi$, $\hat{\mathbf{u}}_{xyz}$ and $r_{\log}$ act as approximations and geometric context for the model.

The simulated BSs were configured with a transmit-power budget of 400 W. Assuming uniform power allocation across the 100 RBs of each BS, we use $P_{\text{ref}} = 4$ W as the reference-CSI acquisition power throughout this work. Features were grouped into CSI ($\text{RI}, \text{CQI}_1, \text{CQI}_2, i_{11}, i_{13}$) and context $\mathcal{C}$ ($i_{\text{RB}}, r_{\log}, \hat{\mathbf{u}}_\phi, \hat{\mathbf{u}}_{xyz}$), with P_{RB} as the decision variable. From there the 5 BS datasets were formed, each as

$$\mathcal{D} = \{(\mathbf{x}, \mathbf{x}', \mathbf{c})\} \quad \mathbf{x} = \text{CSI}_{4W} \quad \mathbf{x}' = \text{CSI}_{P_{\text{RB}}} \quad \mathbf{c} = (P_{\text{RB}}, \mathcal{C}),$$

resulting in 11 million samples and 18 features. To mimic dynamic power allocation, $\mathcal{D}$ was reshaped from $[11 \times 10^6, 18]$ to $[11 \times 10^4, 100, 18]$ by grouping samples into input sequences of 100 CSI spanning all RBs. Within each input sequence, i_{RB} was ordered 0–99, while P_{RB} and $\mathbf{p}_{\text{UE}}$ were randomized. This emulated a BS with 100 RBs of non-uniform power distribution, serving 100 UEs randomly positioned in $\mathcal{S}$. The identity mapping, where $\mathbf{x} = \mathbf{x}'$, was included during training to avoid a discontinuity in the learned discrete mapping T .

C. Training Setup

Each BS dataset was paired with a separate model instance. By making each model depend only on CSI from a single BS, the framework could scale to an arbitrary number of BSs by training additional models on new BS datasets. During training we split inputs along the feature dimension into input-target-condition tuples $(\mathbf{x}, \mathbf{x}', \mathbf{c})$, where $\mathbf{x}$ and $\mathbf{c}$ were provided to the model to generate predictions $\tilde{\mathbf{x}}$, using $\mathbf{x}'$ as the supervision target. Each model was trained for 400 epochs. Over the last 200 epochs, a cosine learning-rate scheduler was applied to periodically anneal the learning rate from 10^{-3} to 10^{-4} , resetting back to 10^{-4} every 10 epochs. Training was repeated across 5 random data shuffles using the seeds $\{2, 4, 8, 16, 32\}$. We used batch size 512, and split the dataset 9:1:1 into a training, validation, and test set after each random data shuffle.

Baseline Models: Three baseline models were introduced to benchmark PDCT. These consisted of an Identity model, a residual ridge-regression model (Ridge), and a 1-nearest-neighbor model (Neighbor), defined as

$$\tilde{\mathbf{x}}_{\text{Identity}} = \mathbf{x}, \quad \tilde{\mathbf{x}}_{\text{Ridge}} = \mathbf{x} + (\mathbf{x} \oplus \mathbf{c})\Theta, \quad \tilde{\mathbf{x}}_{\text{Neighbor}} = \mathbf{x}'_j.$$

For Neighbor, $j = \arg \min_i \|\mathbf{p}_{\text{UE}} - \mathbf{p}_{\text{UE},i}^{\text{train}}\|_2$ among candidate training samples i with i_{RB} and P_{RB} matching the input. These baselines evaluate whether power-domain CSI prediction can be explained by direct CSI reuse, linear feature relationships, or spatial similarity, respectively.

Alternative Losses: Three alternative losses were used to assess PCE. These consisted of cross-entropy (CE), expected ordinal (EO), and earth mover's distance (EMD) [23], as

$$\mathcal{L}_{\text{CE},k} = - \sum_{l=0}^{C_k-1} \mathbf{y}'_{k,l} \log(\tilde{s}_{k,l}), \quad \mathcal{L}_{\text{EO},k} = \left| \sum_{l=0}^{C_k-1} l \tilde{s}_{k,l} - \mathbf{x}'_k \right|, \\ \mathcal{L}_{\text{EMD},k} = \|F(\tilde{s}_k) - F(\mathbf{y}'_k)\|_1.$$

Here, $\mathbf{y}'$ denotes one-hot encoded targets, $\tilde{s}$ predicted softmax probabilities, and F the CDF. These losses provide categorical, ordinal point-estimate, and ordinal distributional baselines for evaluating the proposed PCE loss.

Prediction Metrics: Model performance is reported as the field-wise NMAE given by,

$$\text{NMAE}_k = \frac{\text{MAE}_k}{(C_k - 1)}, \quad \text{MAE}_k = \frac{\|\mathbf{x}'_k - \tilde{\mathbf{x}}_k\|_1}{N}.$$

The maximum absolute error $(C_k - 1)$ normalizes scores to $[0, 1]$, enabling fair field comparisons. NMAE is reported as an average $\pm$ standard deviation over the random data shuffles.

Communication Metrics: Communication performance is reported as the transmission-rate loss l_{rate} , and CSI overestimation rate r_{over} . The transmission rate loss,

$$l_{\text{rate}} = 1 - \frac{\sum_i \tilde{r}_{\text{PHY},i}}{\sum_i r'_{\text{PHY},i}}, \quad r_{\text{PHY}} = \alpha_1 r_{\text{CQI}_1} + \alpha_2 r_{\text{CQI}_2},$$

is the relative reduction in the total PHY-layer transmission rate r_{PHY} for $\tilde{\mathbf{x}}$ with respect to $\mathbf{x}'$. $r_{\text{PHY},i}$ is computed per sample i in $\tilde{\mathbf{x}}$ and $\mathbf{x}'$, with the CQI-to-rate lookup r_{CQI} defined as in [15], and RI multiplicity factors α_1 and α_2 representing the number of active spatial layers associated with each codeword. To avoid rewarding infeasible CSI predictions, we set $\tilde{r}_{\text{PHY},i} = 0$ for any prediction i in $\tilde{\mathbf{x}}$ overestimating the RI, CQI₁, or CQI₂ targets in $\mathbf{x}'$. The overestimation rate,

$$r_{\text{over}} = \frac{1}{N} \sum_i o_i, \quad o_i = \begin{cases} 1, & \text{RI or CQI overestimated,} \\ 0, & \text{otherwise} \end{cases},$$

is the fraction of $\tilde{r}_{\text{PHY},i}$ set to zero for $\tilde{\mathbf{x}}$. Realistically, overestimated RI and CQI values may still contribute to the aggregate throughput, making l_{rate} a conservative estimation.

Experiments: Unless otherwise stated, all experiments used a PDCT instance with 1 encoder block (2.4 M parameters) and the BS₁ dataset. The five experiments we conducted consisted of: (a) evaluating PDCT across the BS₁–BS₅ datasets; (b) evaluating PDCT across sizes of 1–6 encoder blocks; (c) evaluating PDCT and PDCT-FFN against the Identity, Ridge and Neighbor baseline models; (d) evaluating PDCT ablations of either the cosine learning-rate scheduler, MHSA block, or FiLM conditioning; and (e) evaluating the proposed PCE loss against CE, EO, and EMD.

V. NUMERICAL RESULTS

(a) *BS Location and Consistency:* From Fig. 4(a) prediction accuracy is seen to remain consistent across the five BS locations. No deployment exhibits substantially degraded NMAE, suggesting that power-domain CSI prediction is reproducible across the considered BS deployments and that the relative difficulty of the CSI indicators is largely independent of BS location. Cross-BS evaluation was not considered an objective of this work, as changes in BS location alter the geometric conditioning variables and therefore define a different prediction environment.

(b) *Model Size and Scaling:* Fig. 4(b) shows that NMAE decreases monotonically across model sizes of 2.4–11.3 M parameters for all fields except RI, which reaches a minimum at 7.8 M parameters. Overall, the reduction in NMAE diminishes with increasing model size. Table I shows that inference latency per 100 CSI rises from 44 μs to 110 μs across model sizes. Together, these results indicate that small PDCT

instances capture most of the attainable performance, while larger models provide diminishing accuracy gains at increased computational cost. Hardware-wise, reducing FP32 throughput by $3.6\times$ (104.8 to 29.15 TFLOPs) increases inference time by only $2\times$ (44 μs to 88 μs) for the 2.4 M-parameter model.

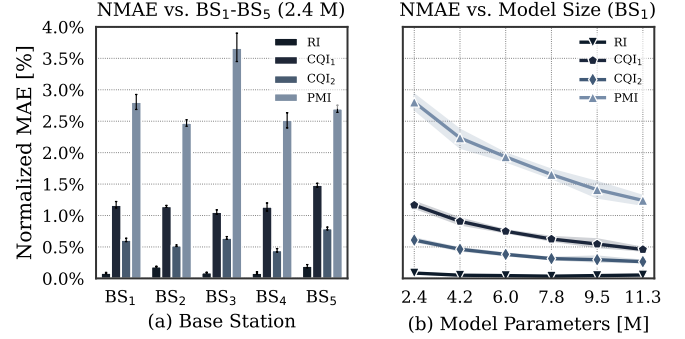


Fig. 4. NMAE across (a) BS₁–BS₅ and (b) increasing model sizes. Performance is largely consistent across BSs and improves with model size.

TABLE I
SIZE AND LATENCY ACROSS PDCT WITH 1-6 ENCODER BLOCKS

Parameters [M]	Memory [MB]	FLOPs [B]	Inference Time [μs]
2.4	9.3	0.5	44.3 ± 1.4
4.2	16.0	0.9	57.2 ± 1.6
6.0	22.8	1.2	69.4 ± 1.4
7.8	29.6	1.6	83.0 ± 1.4
9.5	36.3	2.0	95.7 ± 1.2
11.3	43.1	2.4	110.2 ± 1.7

Inference time is reported per sequence of 100 CSI and averaged over all forward passes during evaluation on an NVIDIA GeForce RTX 5090 GPU.

(c) *Baseline Comparisons:* Table II shows that direct reuse of CSI_{4w} results in large NMAE, rate loss and overestimation. Ridge improves slightly on all metrics except CQI₂ and PMI, while Neighbor improves substantially throughout. Together, these results suggest that power-domain CSI prediction highly benefits from adjacent feature-space structure, and less from residual correction. PDCT and PDCT-FFN employ both and substantially outperform all baselines.

(d) *Architecture Ablation:* As shown in Table II, all three components contribute to performance, with FiLM conditioning having the largest impact. Comparing PDCT-FFN against PDCT isolates the contribution of MHSA without changing the parameter count. While PDCT-FFN achieves slightly lower RI NMAE, PDCT outperforms it on all remaining metrics, indicating that MHSA provides modest but consistent gains beyond model capacity alone. Learning-rate scheduling provides complementary improvements letting the model converge better with finer gradient steps.

(e) *Loss Function Evaluation:* As shown in Table II, PCE (used in PDCT) achieves the lowest NMAE across all CSI fields as well as the lowest rate loss and overestimation. CE remains competitive but consistently underperforms PCE, while EO and EMD substantially degrade both prediction and communication performance. These results indicate that preserving the optimization properties of CE while incorporating ordinal structure is beneficial for CSI prediction.

TABLE II
BASELINE- AND ABLATION COMPARISONS ON DATA FROM BS₁

Group	Architecture	RI [%]	CQI ₁ [%]	CQI ₂ [%]	PMI [%]	Rate Loss [%]	Overestimation Rate [%]
Baselines	Identity	7.07	8.25	11.15	12.49	38.66	37.52
	Ridge	5.33 ± 0.01	7.06 ± 0.01	12.81 ± 0.01	13.94 ± 0.01	37.80	36.07
	Neighbor	2.02 ± 0.01	4.38 ± 0.01	4.06 ± 0.01	9.36 ± 0.01	26.69	23.80
Proposed	PDCT	0.08 ± 0.01	1.17 ± 0.07	0.61 ± 0.03	2.80 ± 0.17	12.59	10.49
	PDCT-FFN	0.07 ± 0.01	1.23 ± 0.03	0.66 ± 0.03	3.02 ± 0.10	13.46	10.72
Ablations	w/o Cos LR	0.22 ± 0.04	1.45 ± 0.07	0.91 ± 0.07	3.59 ± 0.18	15.11	12.40
	w/o MHSA	0.13 ± 0.02	1.54 ± 0.07	0.88 ± 0.08	3.78 ± 0.27	16.94	13.47
	w/o FiLM	0.28 ± 0.03	1.99 ± 0.03	1.29 ± 0.04	5.27 ± 0.09	20.92	17.01
Losses	w/ CE	0.09 ± 0.02	1.47 ± 0.07	0.72 ± 0.05	3.63 ± 0.14	12.83	11.07
	w/ EO	1.12 ± 0.35	2.86 ± 0.22	2.52 ± 0.29	8.25 ± 1.51	30.08	25.12
	w/ EMD	0.95 ± 0.12	10.99 ± 6.17	4.28 ± 2.75	10.76 ± 0.78	32.89	42.93

Results for BS₁. The identity baseline reports the NMAE between CSI_{4W} and the target CSI. PDCT and PDCT-FFN each contain 2.4 M parameters.

VI. CONCLUSION

This paper introduced PDCT and PDCT-FFN for power-domain CSI prediction in downlink MIMO-OFDM systems, together with the PCE loss for ordinal CSI prediction. While existing CSI prediction methods primarily operate across the temporal or spatial domains, the proposed approach predicts CSI feedback indicators across transmit-power configurations. Given CSI acquired at a reference transmit power, the proposed models predict CSI feedback indicators at target transmit-power levels under identical propagation conditions.

The results show that power-domain CSI prediction is feasible across the considered deployment scenarios. Prediction accuracy remained consistent across five BS locations and improved with model size, indicating that the underlying prediction task is reproducible and scalable. Architectural ablations showed that FiLM conditioning contributes the largest performance gains, while comparison between PDCT and PDCT-FFN indicates that MHSA provides smaller but consistent improvements beyond model capacity alone. PCE consistently improved performance over CE, highlighting the importance of incorporating ordinal structure into CSI prediction objectives. Since power allocation requires evaluating CSI under multiple candidate transmit-power configurations, inference latency remains an important design constraint. The measured latency of tens of microseconds per 100 CSI suggests that power-domain CSI prediction is computationally practical for reduced candidate sets or multi-TTI update intervals. Future work will investigate power-domain CSI prediction for multi-BS and multi-scenario power allocation. Robustness to missing data and high-mobility scenarios also remain as future work.

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